

WEEK-7

TASK-1:

Write a program that prints the addresses Of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

CODE:

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

int global_var = 10;      // Global variable

int global_uninitialized;  // BSS segment

void code_function()

{

    printf("Inside code function\n");

}

int main()

{

    static int static_var = 20; // Static/Data segment

    int stack_var = 30;        // Stack variable

    int *heap_var = (int *)malloc(sizeof(int));

    *heap_var = 40;            // Heap variable

    printf("\n===== PROCESS MEMORY ADDRESS SPACE =====\n");

    printf("Code segment address    : %p\n", (void *)code_function);

    printf("Global variable address  : %p\n", (void *)&global_var);

    printf("BSS variable address    : %p\n", (void *)&global_uninitialized);

    printf("Static variable address  : %p\n", (void *)&static_var);

    printf("Heap variable address    : %p\n", (void *)heap_var);

    printf("Stack variable address   : %p\n", (void *)&stack_var);

    printf("\nProcess ID: %d\n", getpid());
```

}

```
Press Enter to exit...
```

```
root@LAPTOP-RJFD8E1T:~# cat /proc/598/maps
62f084468000-62f084461900 r-- 000000000 00:30 13862 /root/memory_layout
62f084461900-62f084462000 r-xp 000010000 00:30 13862 /root/memory_layout
62f084462000-62f084463900 r-- 000020000 00:30 13862 /root/memory_layout
62f084463900-62f084464000 r-- 000020000 00:30 13862 /root/memory_layout
62f084464000-62f084465000 r-- 000030000 00:30 13862 /root/memory_layout
62f02bd36000-62f02bd37000 r-- 000000000 00:00 0 [heap]
7f519330000-7f5193301000 r-- 000220000 00:30 46729 /usr/lib/x86_64-linux-gnu/libc.so.6
7f5193301000-7f5193302000 r-- 000280000 00:30 46729 /usr/lib/x86_64-linux-gnu/libc.so.6
7f5193302000-7f5193304f000 r-- 001160000 00:30 46729 /usr/lib/x86_64-linux-gnu/libc.so.6
7f5193304f000-7f5193305900 r-- 0011f0000 00:30 46729 /usr/lib/x86_64-linux-gnu/libc.so.6
7f5193305900-7f5193306500 r-- 002020000 00:30 46729 /usr/lib/x86_64-linux-gnu/libc.so.6
7f5193306500-7f5193312000 r-- 000000000 00:00 0 [var]
7f5193304000-7f5193307000 r-- 000000000 00:00 0 [var,vclock]
7f519330e000-7f519330e000 r-- 000000000 00:00 0 [vdso]
7f5193306000-7f5193307000 r-- 000000000 00:30 46726 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7f5193307000-7f5193309c00 r-xp 000010000 00:30 46726 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7f5193309c00-7f519330cc00 r-- 0002c0000 00:30 46726 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7f519330cc00-7f519330ce00 r-- 000336000 00:30 46726 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7f519330ce00-7f519330d000 r-- 000338000 00:30 46726 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7fff795a3800-7fff7952a000 r-- 000000000 00:00 0 [stack]
root@LAPTOP-RJFD8E1T:~# cat /proc/598/maps | grep -E '\[heap\]|\[stack\]|memory_layout'
62f084468000-62f084461900 r-- 000000000 00:30 13862 /root/memory_layout
62f084461900-62f084462000 r-xp 000010000 00:30 13862 /root/memory_layout
62f084462000-62f084463900 r-- 000020000 00:30 13862 /root/memory_layout
62f084463900-62f084464000 r-- 000020000 00:30 13862 /root/memory_layout
62f084464000-62f084465000 r-- 000030000 00:30 13862 /root/memory_layout
62f02bd36000-62f02bd37000 r-- 000000000 00:00 0 [heap]
7fff795a3800-7fff7952a000 r-- 000000000 00:00 0 [stack]
root@LAPTOP-RJFD8E1T:~#
```

Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
Press Enter to exit...
```

[illegible]

